

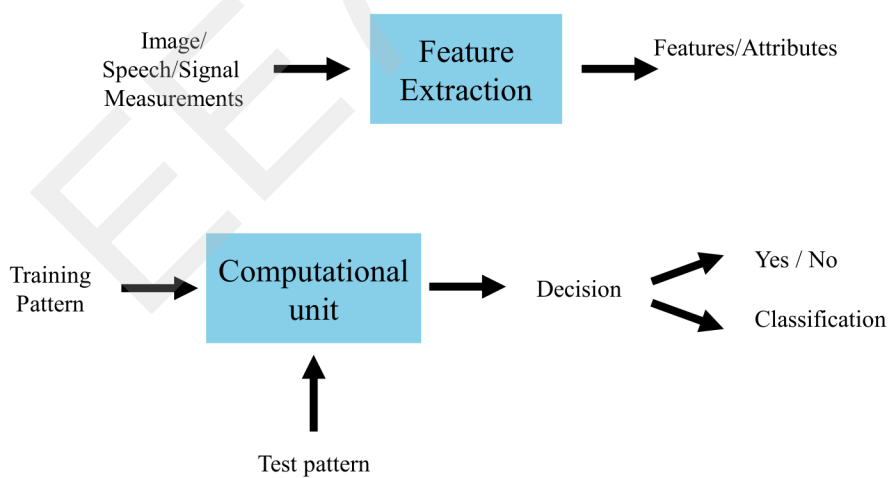
EE708: Fundamentals of Data Science and Machine Intelligence

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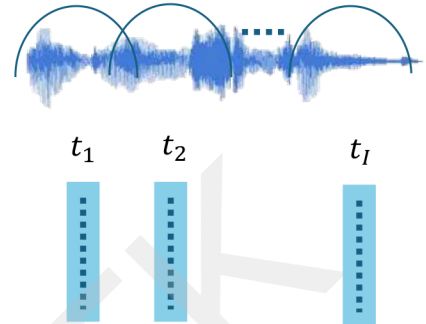
Module 4A Clustering

Pattern Recognition



Pattern Recognition

- How to compare patterns?
- Test pattern $\tau = \{t_1, t_2, \dots, t_I\}$
- Where t_i is the spectral vector at time i and I is the total number of frames (in the case of long duration signal like speech and other time series)
- Reference patterns $R^j = \{r_1^j, r_2^j, \dots, r_j^j\}$
- Objective of pattern comparison: Evaluate distance of τ to each R^j for $1 \leq j \leq V$



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Distance Metric/Measure

- Assume we have two test vectors x and y , defined on a vector space χ ,
- A distance metric d , on the vector space χ , as a real valued function on the cartesian product $\chi \times \chi$, is defined as follows
 - $0 \leq d(x, y) < \infty$, for $x, y \in \chi$ and $d(x, y) = 0$, iff $x = y$ → Positive definite
 - $d(x, y) = d(y, x)$, for $x, y \in \chi$ → Symmetry
 - $d(x, y) \leq d(x, z) + d(y, z)$, for $x, y \in \chi$ → Triangle inequality
- Distance function is invariant if

$$d(x + z, y + z) = d(x, y)$$
- Distortion measure should satisfy positive definiteness.

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Distance Metric/Measure

Euclidean Distance:

- Let $x = [x_1, x_2]$ and $y = [y_1, y_2]$ two vector in 2-D space

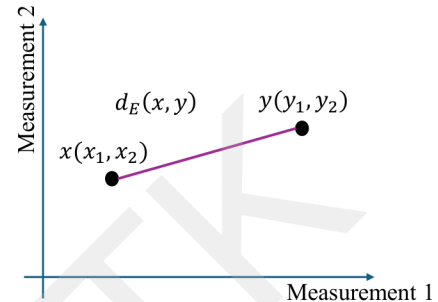
- The Euclidean distance is

$$d_E(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

- For N-Dimensional space

$$d_E(x, y) = \sqrt{\sum_{i=1}^N (x_i - y_i)^2} = \|\mathbf{x} - \mathbf{y}\|_2$$

- Also called L_2 norm (L_2 distance)
- The measure is **invariant to rotation and translation.**



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Distance Metric/Measure

Manhattan Distance:

- Let $x = [x_1, x_2]$ and $y = [y_1, y_2]$ two vector in 2-D space

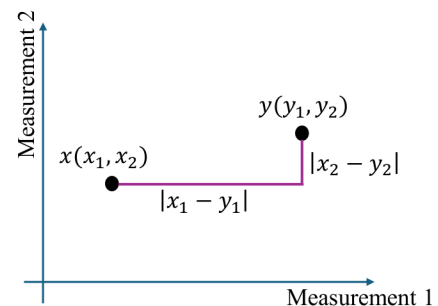
- The Manhattan distance is

$$d_M(x, y) = |x_1 - y_1| + |x_2 - y_2|$$

- For N-Dimensional space

$$d_E(x, y) = \sum_{i=1}^N |x_i - y_i| = \|\mathbf{x} - \mathbf{y}\|_1$$

- Also called L_1 norm (L_1 distance)
- The measure is **invariant to translation only.**



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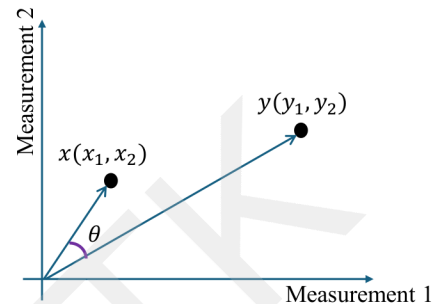
Distance Metric/Measure

Cosine Distance:

- Let $x = [x_1, x_2]$ and $y = [y_1, y_2]$ two vector in 2-D space
- The cosine distance measures the angle between the two vectors x and y .

$$d_c(x, y) = \theta = \cos^{-1} \frac{x \cdot y}{\|x\| \|y\|}$$

- The measure **is invariant to rotation**.



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Query

"neural information retrieval"

Prompt = explain rotational invariance of cos distance with document retrieval example; give a better examples where cosine distance beats others

Two documents (same topic words, different lengths)

Doc A (short, focused)

"Neural information retrieval uses embeddings and dense vectors for search."

Doc B (very long, same core topic but lots of extra content)

"Neural information retrieval uses embeddings and dense vectors for search...
(plus many extra paragraphs about unrelated background, history of AI, datasets, company news, long introductions, etc.)"

Both contain the query terms, but **Doc B** has **any more total words**, so its vector often has **larger magnitude**.

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1) Beats dot product when a long/off-topic doc has huge magnitude

Feature space: [retrieval, embeddings, unrelated]

- Query $q = [1, 1, 0]$
- Doc A (short + on-topic) $d_A = [1, 1, 0]$
- Doc B (lots of extra unrelated text) $d_B = [10, 10, 100]$

Dot product (can be fooled by magnitude)

- $q \cdot d_A = 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 = 2$
- $q \cdot d_B = 1 \cdot 10 + 1 \cdot 10 + 0 \cdot 100 = 20$

↓

Dot product ranks Doc B higher ($20 > 2$), even though it's bloated with unrelated content.

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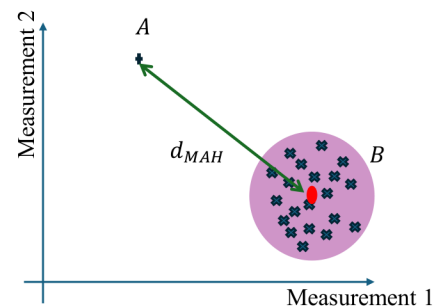
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Distance Metric/Measure

Mahalanobis Distance:

- Measure of the distance between a point A and a distribution B .
- It measures the cluster spacing as well as the cluster spread.
- Let $B \in \mathbb{R}^N$, has a probability distribution with mean μ and positive semi-definite covariance Σ .
- The distance of point A from B is

$$d_{MAH} = \sqrt{(A - \mu)\Sigma^{-1}(A - \mu)}$$



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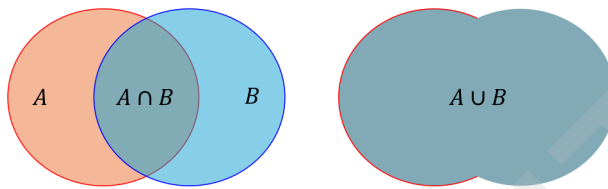
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Distance Metric/Measure

Jaccard Distance/Index/Coefficient:

- The Jaccard distance measures the similarity of the two data set
- Let A and B be two datasets, then the Jaccard Index is

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$



$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

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Distance Metric/Measure

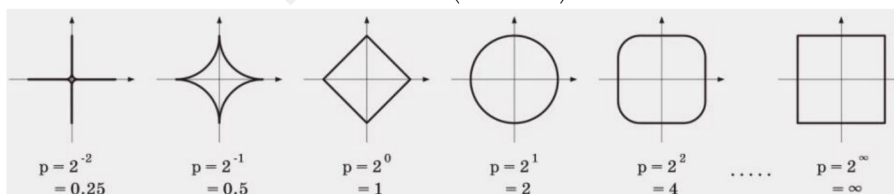
Minkowski Distance: Euclidean, Manhattan, Chebyshev Distances – Unified Generalization

$$d(x, y) = \left(\sum_{i=1}^K |x_i - y_i|^p \right)^{\frac{1}{p}}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_K \end{bmatrix}, y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_K \end{bmatrix}$$

- $p \geq 1$ Minkowski distance is a metric, for $p < 1$ Minkowski distance is not a metric

$$\begin{array}{ll} \text{Manhattan} & \text{Euclidean} \\ \sum_{i=1}^K |x_i - y_i| & \left(\sum_{i=1}^K |x_i - y_i|^2 \right)^{\frac{1}{2}} \end{array}$$

$$\begin{array}{l} \text{Chebyshev Distance } (p = -\infty, \infty) \\ \lim_{p \rightarrow -\infty} \left(\sum_{i=1}^K |x_i - y_i|^p \right)^{\frac{1}{p}} = \min_{i=1, \dots, n} |x_i - y_i|, \\ \lim_{p \rightarrow \infty} \left(\sum_{i=1}^K |x_i - y_i|^p \right)^{\frac{1}{p}} = \max_{i=1, \dots, n} |x_i - y_i| \end{array}$$

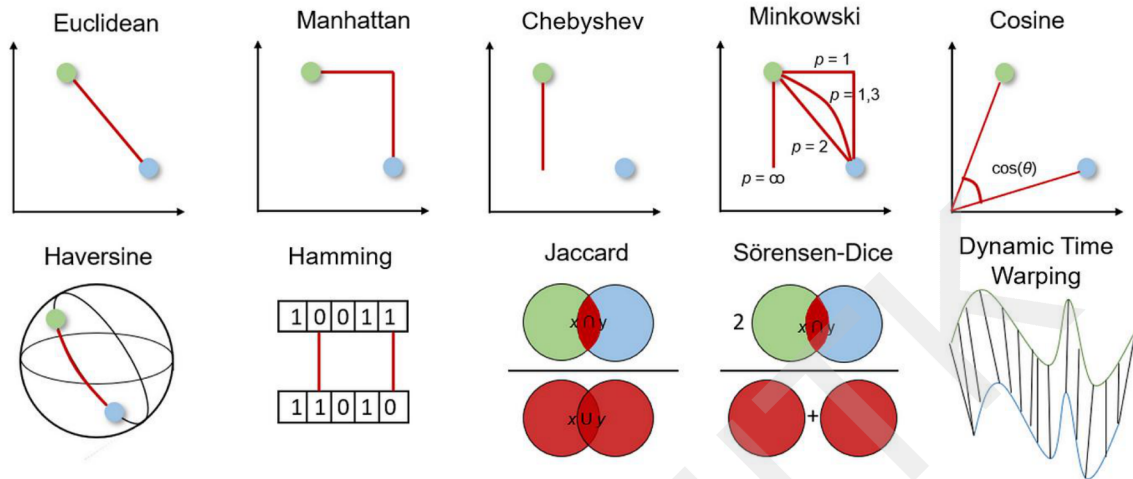


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Distance Metric/Measure



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Distance Metric/Measure

Spectral Distortion Measures

- Let $S(w)$ and $S'(w)$ be two spectra and $d(S, S')$ be the distance measure
- The Log spectral distance (LSD) is

$$V(w) = \log S(w) - \log S'(w)$$

- In general

- $d(S, S')^p = (d_p)^p = \int_{-\pi}^{\pi} |V(w)|^p \frac{d\omega}{2\pi}$
- $d(S, S')^p$ defines a set of L_p norms where

$p = 1$	Mean absolute LSD
$p = 2$	RMS LSD
$p = \infty$	Peak LSD

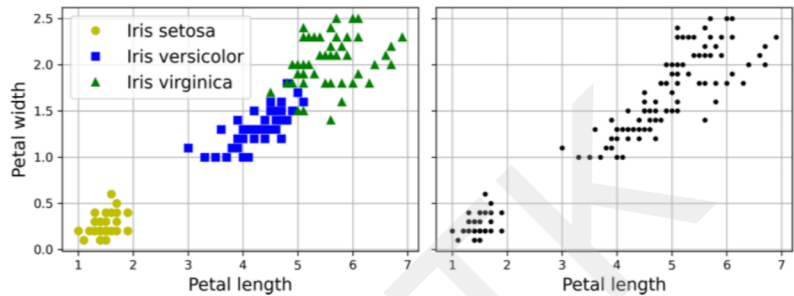
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Clustering

- Task of identifying similar instances and assigning them to clusters, or groups of similar instances.
- Clustering is used in a wide variety of applications, including:
 - Customer segmentation
 - Data analysis
 - Dimensionality reduction
 - Feature engineering
 - Anomaly detection
 - Semi-supervised learning
 - Search engines
 - Image segmentation



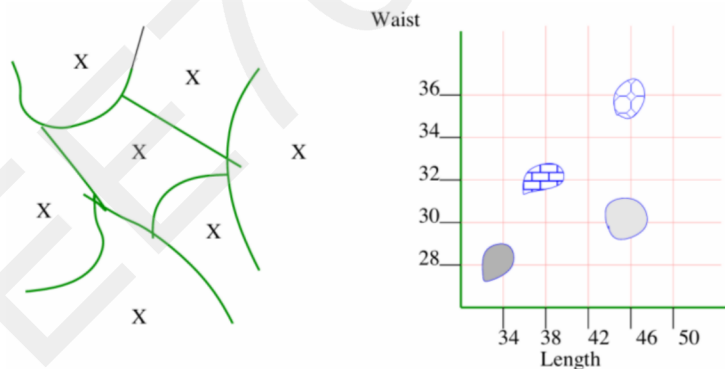
Classification (left) versus clustering (right)

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Clustering: Readymade Trouser Example



Partitioned vector space
X is the centroid of region

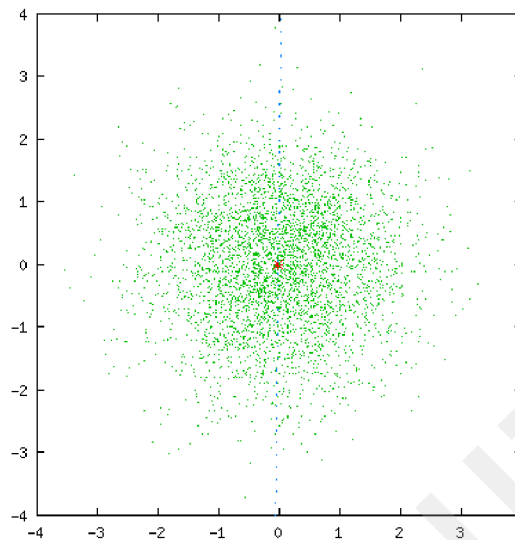
Clusters in pattern space

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Clustering: Animation



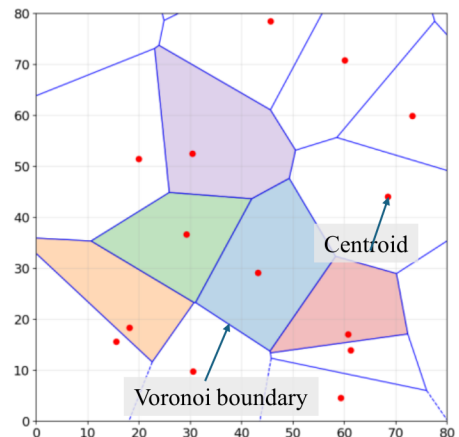
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Clustering: K-means Algorithm (Lloyds Algorithm)

- Objective: Set of N training vectors needs to be clustered into a set of M codebook vectors.
- 1. Initialization: Arbitrarily select M vectors out of N vectors. This will yield initial codebook.
- 2. Nearest neighbor search: For each training vector find codeword in the current codebook that is closest and assign that to the current cell.
- 3. Centroid update: Update the code word of each cell using centroid of the training vectors in that cell.
- 4. Iteration: Repeat 2 and 3 till convergence. Convergence criterion can be:
 - i. Average distance falls below a certain threshold
 - ii. Number of cross cell migrations are below a threshold
 - iii. Fixed number of iterations



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Clustering: Binary Split Vector Clustering/Quantization (BSVQ)

1. Design a 1 vector codebook
 2. Double the size of the codebook. Split codebook according to a rule

$$y_n^+ = y_n(1 + \epsilon)$$

$$y_n^- = y_n(1 - \epsilon)$$

n goes from 1 to current size of the codebook and ϵ is the splitting parameter $0.01 \leq \epsilon \leq 0.05$
 3. Use the K-means algorithm to get best of centroids for the split codebook
 4. Iterate step 2 and 3 until a codebook of size M is designed.
- Mathematically BSVQ clustering can be done in two ways:
 - i. Reducing average distortion
 - ii. Minimizing a cost function

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Clustering: Binary Split Vector Clustering/Quantization (BSVQ)

- i. Reducing average distortion
 - VQ tries to reduce average distortion in each cell of the vector space
 - The average distortion in cell C_i is given by

$$D_i = \frac{1}{N} \sum_{x \in C_i} d(x, z_i)$$

where z_i is the centroid of the cell C_i and $d(x, z_i) = (x - z_i)^T(x - z_i)$

- The final centroid obtained (after convergence) are then stored in a codebook.

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Clustering: Binary Split Vector Clustering/Quantization (BSVQ)

ii. Minimizing a cost function

- Objective function (distortion measure) can be defined as

$$J = \sum_{n=1}^N \sum_{k=1}^K \|x_n - \mu_k\|^2 r_{nk}$$

Which represents sum of square of the distance of each feature data point to its assigned mean feature vector μ_k , where $r_{nk} \in \{0,1\}$ is binary indicator variable.

- To minimize J , two stage minimization can be done.

Objective: Minimize J to find r_{nk} and μ_k

Stage 1: Keep μ_k fixed, find r_{nk} by minimizing J

Stage 2: Keet r_{nk} fixed, find μ_k by minimizing J

Clustering: Binary Split Vector Clustering/Quantization (BSVQ)

- Clustering to the closest center

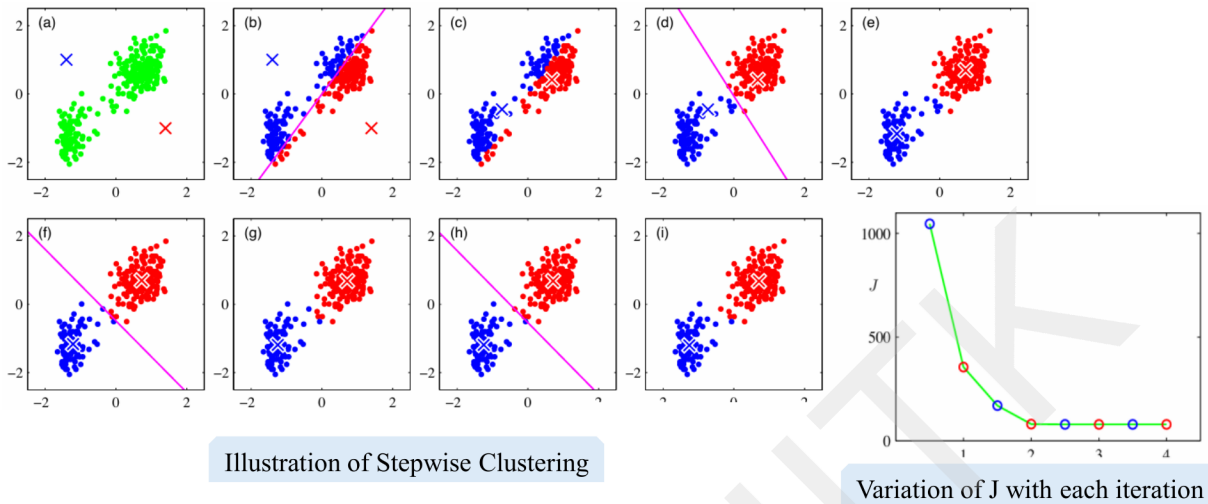
$$r_{nk} = \begin{cases} 1, & k = \arg \min_j \|x_n - \mu_j\|^2 \\ 0, & \text{otherwise} \end{cases}$$

- Minimizing J : Setting the derivative of J with respect to μ_k to zero

$$2 \sum_{n=1}^N r_{nk} (x_n - \mu_k) = 0 \Rightarrow \mu_k = \frac{\sum_{n=1}^N x_n r_{nk}}{\sum_{n=1}^N r_{nk}}$$

- μ_k is the mean of all data points assigned to the cluster k

Clustering: Binary Split Vector Clustering/Quantization (BSVQ)



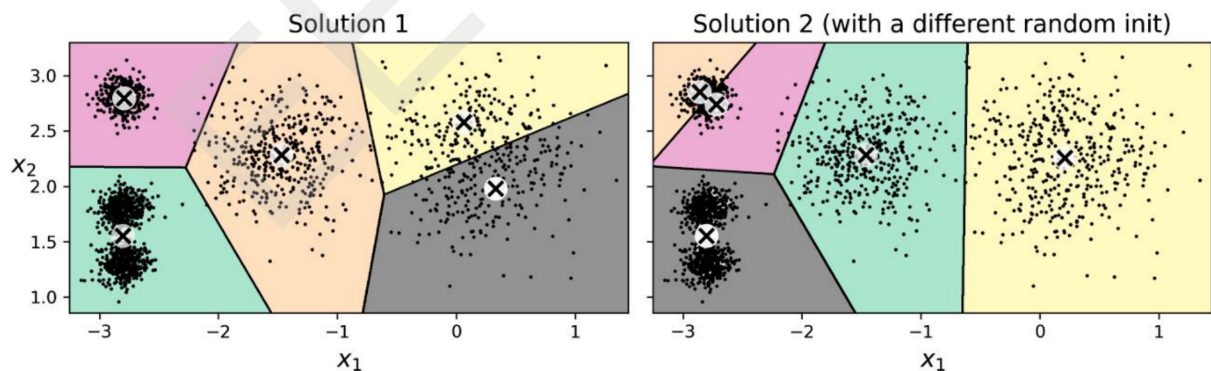
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Significance of Centroid Initialization

- Although the algorithm is guaranteed to converge, it may not converge to the right solution (i.e., it may converge to a local optimum).
- Whether it does or not depends on the centroid initialization.



Suboptimal solutions due to unlucky centroid initializations.

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Online Clustering

- Online Stochastic Algorithm (Sequence update)

$$\underbrace{\mu_k^{new}}_{\text{Adapted new mean}} = \underbrace{\mu_k^{old}}_{\text{old mean}} + \underbrace{h_n}_{\substack{\text{learning} \\ \text{parameter weight}}} \underbrace{(x_n - \mu_k^{old})}_{\text{distance}}$$

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K-means Variants: Accelerated K-means and Mini-batch K-means

- Accelerated k-means (*Charles Elkan, 2003*):
 - On some large datasets with many clusters, the algorithm can be accelerated by avoiding many unnecessary distance calculations.
 - Exploits triangle inequality and keeps track of lower and upper bounds for distances between instances and centroids.
 - Does not always accelerate training, and sometimes it can even slow down training significantly.
 - Depends on the dataset.
- Mini-batch k-means (*David Sculley, 2010*):
 - Instead of using the full dataset at each iteration, the algorithm uses mini-batches, moving the centroids just slightly at each iteration.
 - Speeds up the algorithm and makes it possible to cluster huge datasets that do not fit in memory.

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Accelerated K-means

- Accelerating k-means approach is based on the triangle inequality.
 - For any three points x , y , and z

$$d(x, z) \leq d(x, y) + d(y, z)$$
- Suppose:
 - $u(x) \geq d(x, c)$ is an upper bound on the distance between x and the center c to which x is currently assigned
 - $l(x, c') \leq d(x, c')$ is a lower bound on the distance between x and some other center c'
- If $u(x) \leq l(x, c')$ then $d(x, c) \leq u(x) \leq l(x, c') \leq d(x, c')$, so it is not necessary to calculate either $d(x, c)$ or $d(x, c')$.
- Note that it will never be necessary in **this iteration** of the accelerated method to compute $d(x, c')$, but it may be necessary to compute $d(x, c)$ exactly because of some other center c'' for which $u(x) \leq l(x, c'')$ is not true.

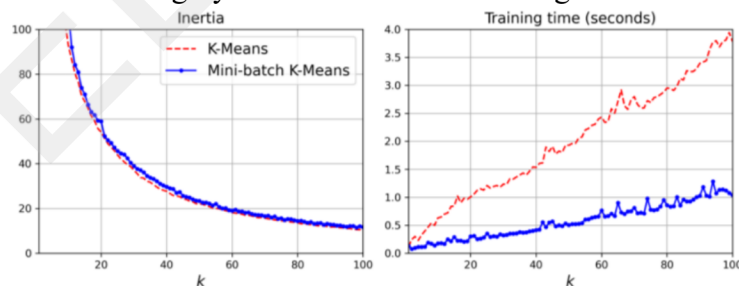
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K-means Variants: Mini-batch k-means

- The difference between the two curves is small but visible.
- Mini-batch k-means have higher inertia ($\sum_{n=1}^N \min_{\mu_j \in C} (\|x_n - \mu_j\|^2)$) than k-means as k increases.
- Mini-batch k-means are roughly 3.5 times faster than regular k-means for the five-blob dataset.



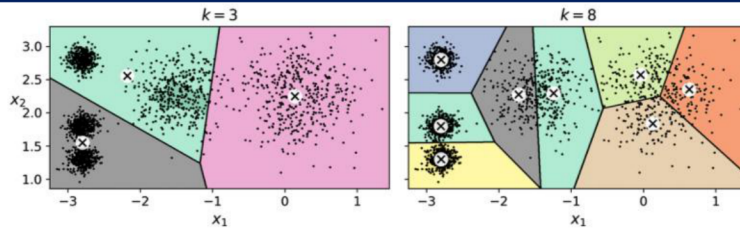
Comparison of inertia and Training time of mini-batch k-means and regular k-means models trained on a five-blob dataset using various numbers of clusters k .

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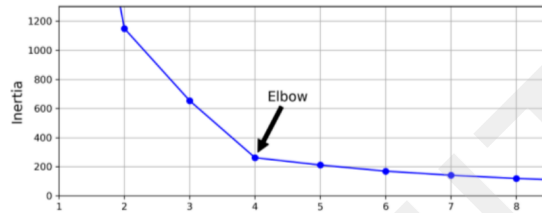
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Optimal Number of Clusters



Bad choices for the number of clusters: when k is too small, separate clusters get merged (left), and when k is too large, some clusters get chopped into multiple pieces (right)



Plotting the inertia as a function of the number of clusters k

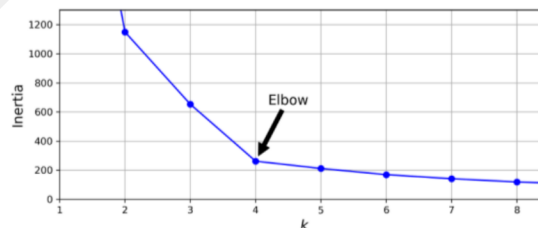
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Optimal Number of Clusters

- The number of clusters (k) is set to $k=5$ by looking at the data.
- In general, it isn't easy to set k , and the result might be bad if it is set to the wrong value.
- Picking the model with the lowest inertia is also not a good choice.
 - The inertia is not a good performance metric when choosing k because it keeps getting lower as k increases.
 - Indeed, the more clusters there are, the closer each instance will be to its closest centroid and, therefore, the lower the inertia.
- The inflection point in the plot of inertia as a function of k is called the *elbow*.



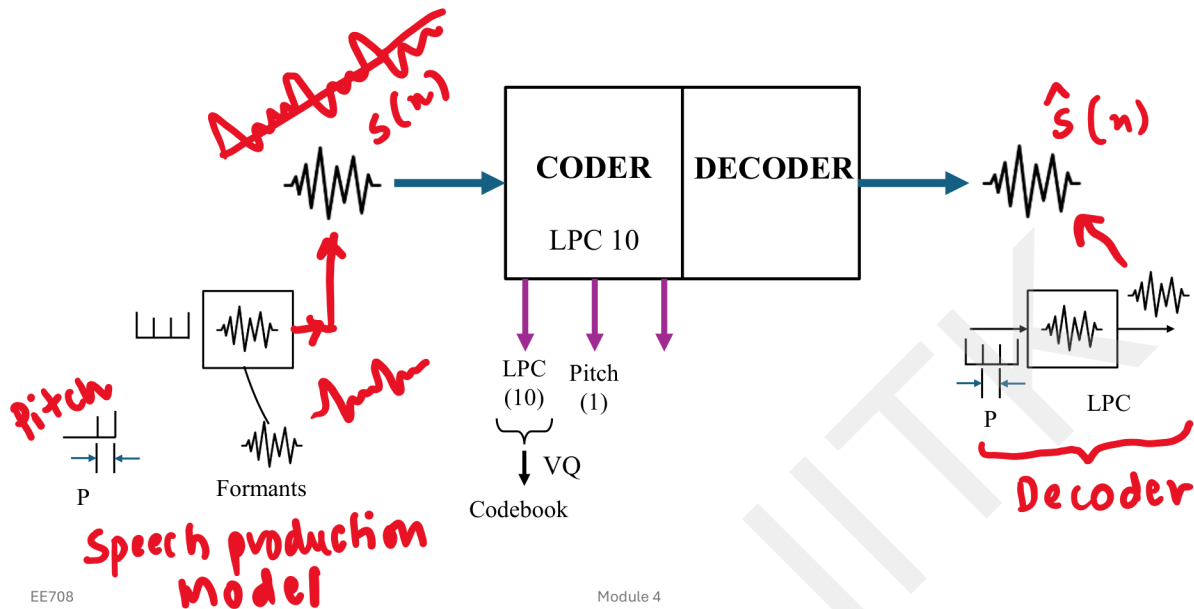
Plotting the inertia as a function of the number of clusters k

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Case Study 1 : Clustering/VQ in Audio Codec Realization



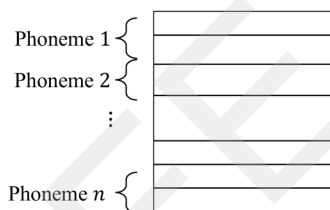
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Audio Codes : Vector Classification Procedure

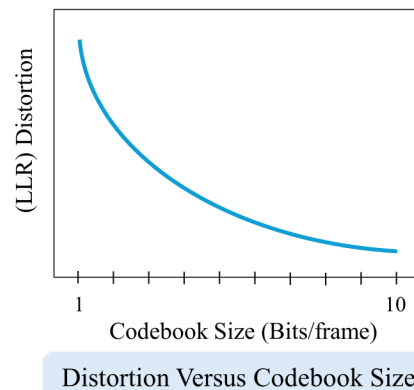
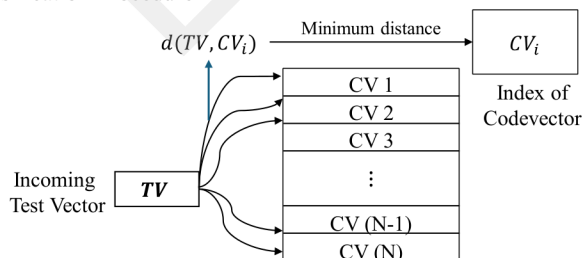
VQ Codebook



$$m^* = \arg \min_{1 \leq m \leq M} d(v, y_m)$$

v is test vector
 y_m is code vector

Vector Classification Procedure

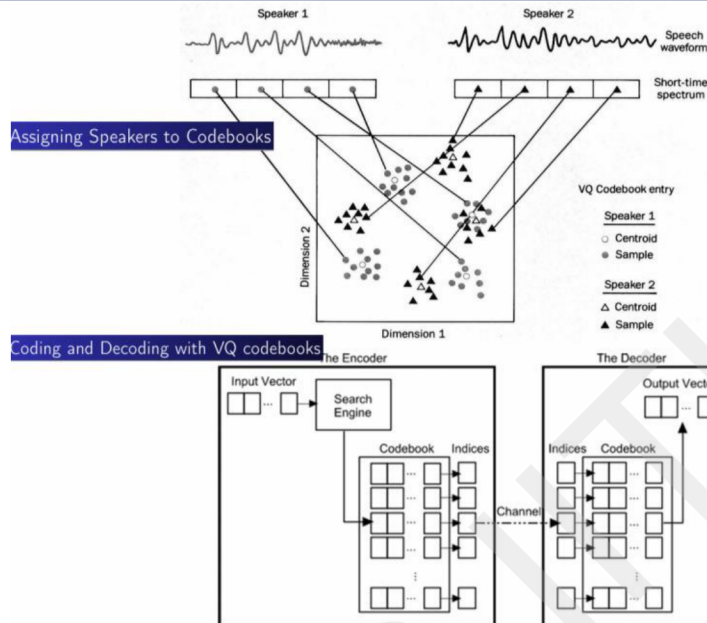


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Audio Codecs : Vector Classification Procedure



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Case Study 2: Clustering for Image Segmentation

- Image segmentation is the task of partitioning an image into multiple segments. There are several variants:
 - Color segmentation
 - Pixels with a similar color get assigned to the same segment.
 - For example, to analyze satellite images to measure how much total forest area there is in a region, color segmentation may be just fine.
 - Semantic segmentation
 - All pixels that are part of the same object type get assigned to the same segment.
 - For example, in a self-driving car's vision system, all pixels that are part of a pedestrian's image might be assigned to the "pedestrian" segment (there would be one segment containing all the pedestrians).
 - Instance segmentation
 - All pixels that are part of the same individual object are assigned to the same segment.
 - In this case there would be a different segment for each pedestrian.

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Clustering for Image Segmentation: Channels

- The image is represented as a 3D array.
- The first dimension's size is the height; the second is the width; and the third is the number of color channels, in this case, red, green, and blue (RGB).
- For each pixel there is a 3D vector containing the intensities of red, green, and blue as unsigned 8-bit integers between 0 and 255.
- Some images may have fewer channels (such as grayscale images, which only have one)
- Some images may have more channels (such as images with an additional alpha channel for transparency or satellite images, which often contain channels for additional light frequencies (like infrared)).

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Clustering for Image Segmentation: Flower with Ladybug

Experiment with various numbers of clusters.

- When fewer than eight clusters, the ladybug's flashy red color fails to get a cluster of its own, it merges colors from the environment.
- This is because k-means prefer clusters of similar sizes.
- The ladybug is small-much smaller than the rest of the image.
- Even though its color is flashy, k-means fails to dedicate a cluster to it.

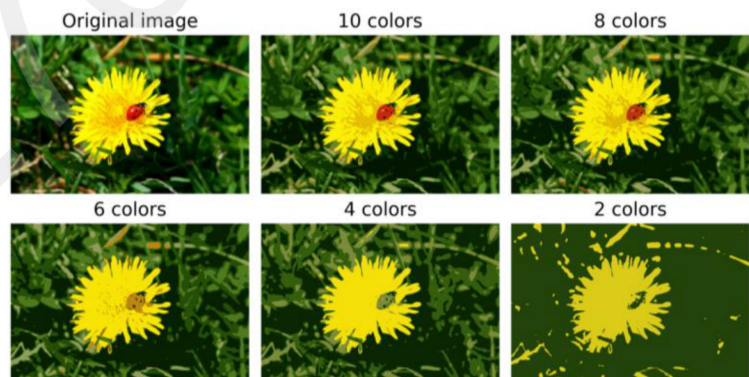


Image segmentation using k-means with various numbers of color clusters

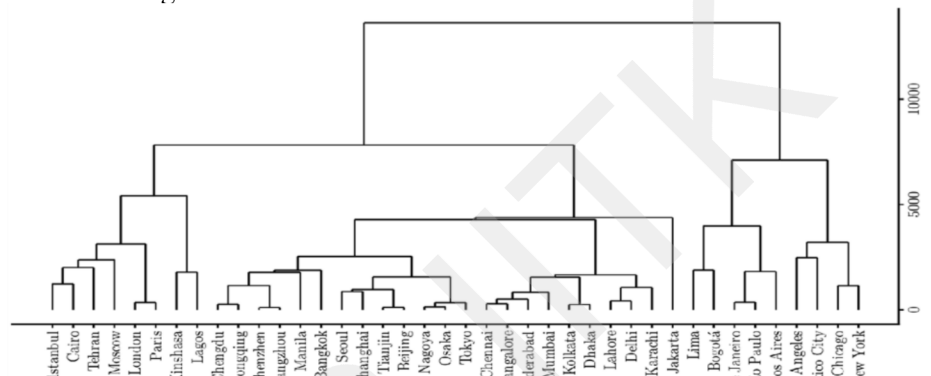
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Hierarchical Clustering

- Involves creating clusters that have a predominant ordering from top to bottom.
- Main advantage of hierarchical clustering is that it does not need to specify the number of clusters; the model determines that by itself.
- This clustering technique is divided into two types:
 1. **Agglomerative** hierarchical clustering
 2. **Divisive** hierarchical clustering



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Hierarchical Clustering: Agglomerative Hierarchical Clustering

1. Agglomerative hierarchical clustering
 - Most common type of hierarchical clustering
 - Used to group objects based on their similarity.
 - It is a **“bottom-up”** approach where each observation starts in its own cluster, and pairs of clusters are merged as one moves up the hierarchy.
 - Delivers a local optimum.
 - Proceeds as follows:
 - i. Make each data point a single-point cluster and form K clusters.
 - ii. Take the two closest data points and combine them, leaving $K - 1$ clusters.
 - iii. Take the two closest clusters and combine them, forming $K - 2$ clusters.
 - iv. Repeat step iii until left with only one cluster.

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Hierarchical Clustering: Divisive Hierarchical Clustering

2. Divisive hierarchical clustering

- It is a **“top-down”** and sequentially splits the remaining clusters to produce the most distinct subgroups.
- **Both A and D**, produce **$K - 1$ hierarchical levels** and facilitate the clustering creation at the level that best partitions data into homogeneous groups.

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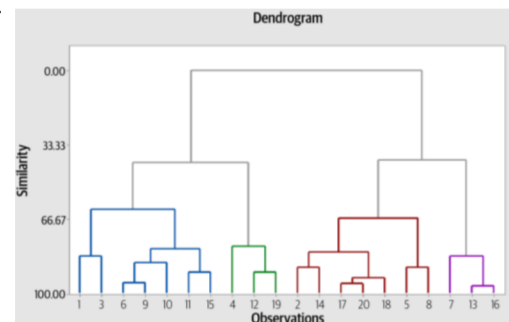
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Hierarchical Clustering : Dendrograms

- Hierarchical clustering enables the plotting of **dendrograms**, which are visualizations of a binary hierarchical clustering.

- Dendrogram:

- A type of tree diagram showing hierarchical relationships between different sets of data.
- Provide an interesting and informative visualization of hierarchical clustering results.
- Contains the memory of the hierarchical clustering algorithm, cluster formation can be obtained by simply inspecting the chart.



Hierarchical clustering

- The **distance between data points** represents **dissimilarities**, and the **height of the blocks** represents the **distance between clusters**.
- Observations that fuse at the bottom are similar, while those at the top are quite different.

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Hierarchical Clustering

- With dendrograms, conclusions are made based on the location of the vertical axis rather than on the horizontal one.
- The advantages of hierarchical clustering are that it is **easy to implement** it, does not require one to specify the number of clusters.
- However, the time complexity for hierarchical clustering can result in **long computation times** relative to other algorithms, such as k-means.
- For a **large dataset**, it can be **difficult to determine the correct number of clusters** by looking at the dendrogram.
- Hierarchical clustering is very **sensitive to outliers**, and in their presence, model performance decreases significantly.

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Hierarchical Cluster Analysis: Algorithm

```

initialize  $t := 0$ ,  $S^{(0)} := \{S_1^{(0)}, \dots, S_N^{(0)}\} = \{\{x_1\}, \dots, \{x_N\}\}$ 
while  $|S^{(t)}| > 1$  do
    update  $t \leftarrow t + 1$ , initialize  $S^{(t)} := S^{(t-1)}$ 
    # Determine cluster with smallest distance:
    for  $i = 2$  to  $|S^{(t)}|$  do
        for  $j = 1$  to  $i - 1$  do
            compute  $\Delta_{ij} := D(S_i^{(t)}, S_j^{(t)})$ 
        end
    end
    determine  $i, j$  with smallest distance  $\Delta_{ij}$ 
    # Merge clusters with smallest distance:
    update  $S_i^{(t)} \leftarrow S_i^{(t)} \cup S_j^{(t)}$ , drop  $S_j^{(t)}$ 
end

```

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Hierarchical Cluster Analysis: Numerical Example

- Let a dataset $X = \{1, 2, 5, 8, 10\}$
- Euclidean distance as similarity metric.
- At $t = 0$:

$$S_i^{(0)} = \{x_i\}, \quad i = 1, 2, \dots, 5$$

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Hierarchical Cluster Analysis: Numerical Example

At $t = 1$:

Step 1: Initial clusters	$S^{(1)} = \left\{ \underbrace{\{1\}}_{S_1^{(1)}}, \underbrace{\{2\}}_{S_2^{(1)}}, \underbrace{\{5\}}_{S_3^{(1)}}, \underbrace{\{8\}}_{S_4^{(1)}}, \underbrace{\{10\}}_{S_5^{(1)}} \right\}$
Step 2: Compute Δ	$\Delta = \begin{bmatrix} 0 & 1 & 4 & 7 & 9 \\ 1 & 0 & 3 & 6 & 8 \\ 4 & 3 & 0 & 3 & 5 \\ 7 & 6 & 3 & 0 & 2 \\ 9 & 8 & 5 & 2 & 0 \end{bmatrix}$
Step 3: Smallest distance between two sets	1
Sets corresponding to smallest distance	$S_1^{(1)}$ and $S_2^{(1)}$
Step 4: Updated clusters	$\{1, 2\}, \{5\}, \{8\}, \{10\}$

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Hierarchical Cluster Analysis: Numerical Example

At $t = 2$:

Step 1: Initial clusters	$S^{(2)} = \left\{ \underbrace{\{1,2\}}_{S_1^{(2)}}, \underbrace{\{5\}}_{S_2^{(2)}}, \underbrace{\{8\}}_{S_3^{(2)}}, \underbrace{\{10\}}_{S_4^{(2)}} \right\}$
Step 2: Compute Δ	$\Delta = \begin{bmatrix} 0 & 3.5 & 6.5 & 8.5 \\ 3.5 & 0 & 3 & 5 \\ 6.5 & 3 & 0 & 2 \\ 8.5 & 5 & 2 & 0 \end{bmatrix}$
Step 3: Smallest distance between two sets	2
Sets corresponding to smallest distance	$S_3^{(2)}$ and $S_4^{(2)}$
Step 4: Updated clusters	$\{1,2\}, \{5\}, \{8,10\}$

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Hierarchical Cluster Analysis: Numerical Example

At $t = 3$:

Step 1: Initial clusters	$S^{(3)} = \left\{ \underbrace{\{1,2\}}_{S_1^{(3)}}, \underbrace{\{5\}}_{S_2^{(3)}}, \underbrace{\{8,10\}}_{S_3^{(3)}} \right\}$
Step 2: Compute Δ	$\Delta = \begin{bmatrix} 0 & 3.5 & 7.5 \\ 3.5 & 0 & 4 \\ 7.5 & 4 & 0 \end{bmatrix}$
Step 3: Smallest distance between two sets	3.5
Sets corresponding to smallest distance	$S_1^{(3)}$ and $S_2^{(3)}$
Step 4: Updated clusters	$\{1,2,5\}, \{8,10\}$

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Hierarchical Cluster Analysis: Numerical Example

At $t = 4$:

Step 1: Initial clusters	$S^{(4)} = \left\{ \underbrace{\{1,2,5\}}_{S_1^{(4)}}, \underbrace{\{8,10\}}_{S_2^{(4)}} \right\}$
Step 2: Compute Δ	$\Delta = \begin{bmatrix} 0 & 6.33 \\ 6.33 & 0 \end{bmatrix}$
Step 3: Smallest distance between two sets	6.33
Sets corresponding to smallest distance	$S_1^{(4)}$ and $S_2^{(4)}$
Step 4: Updated clusters	$\{1,2,5,8,10\}$

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Hierarchical Cluster Analysis: Numerical Example

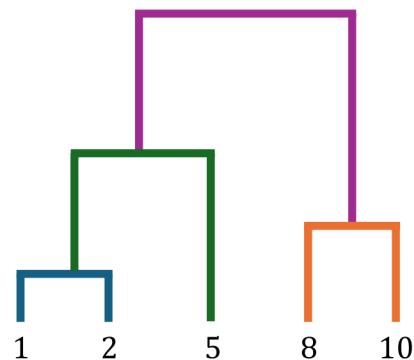
$t = 4: \{1,2,5,8,10\}$

$t = 3: \{1,2,5\}, \{8,10\}$

$t = 2: \{1,2\}, \{5\}, \{8,10\}$

$t = 1: \{1,2\}, \{5\}, \{8\}, \{10\}$

$t = 0: \{\{1\}, \{2\}, \{5\}, \{8\}, \{10\}\}$



Dendrogram for the numerical example

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Hierarchical Cluster Analysis: Inter Cluster Distance

- Let $\{x_1, x_2, \dots, x_N\}$ be the data points that needs to be grouped into clusters.
- Given a distance measure $\delta(\cdot, \cdot)$ to compare those data points
- The distance $D(A, B)$ between two arbitrary clusters of data points A and B , can be chosen from

$$D_{min}(A, B) = \min_{u \in A, v \in B} \{\delta(u, v)\}$$

$$D_{avg}(A, B) = \langle \delta(u, v) \rangle_{u \in A, v \in B}$$

$$D_{max}(A, B) = \max_{u \in A, v \in B} \{\delta(u, v)\}$$

Hierarchical Cluster Analysis

- Depending on whether D_{min} , D_{avg} or D_{max} is used as the distance measure between clusters, we speak of **single-linkage**, **average-linkage**, or **complete-linkage clustering**.
- A divisive hierarchical cluster analysis continuously refines partitions, **starting from a single cluster** that covers the whole dataset.
- In a hierarchy of clusterings, every level of the hierarchy represents a different partition.
- **Hierarchies can also be called rooted trees.**
- Every vertex of the tree represents a cluster, starting at the root, which represents the largest possible cluster-the cluster which contains all of the data points.
- With each level in the direction of the tree's leaves, the partitions become finer and finer, containing smaller and smaller clusters.
- Recollect : Dendrogram can be used to visualize hierarchy/tree.

Hierarchical Cluster Analysis: Example City Distances

- An example that represents the **clustering of forty international cities by geo-graphical distance** using the average-linkage method.
- The leaves correspond to the cities to be grouped, and each bifurcation represents the merger of two clusters of cities.
- The position of the bifurcation can be used to read off the value of the cluster distance function $D_{avg}(\cdot, \cdot)$ at which the merger occurred.

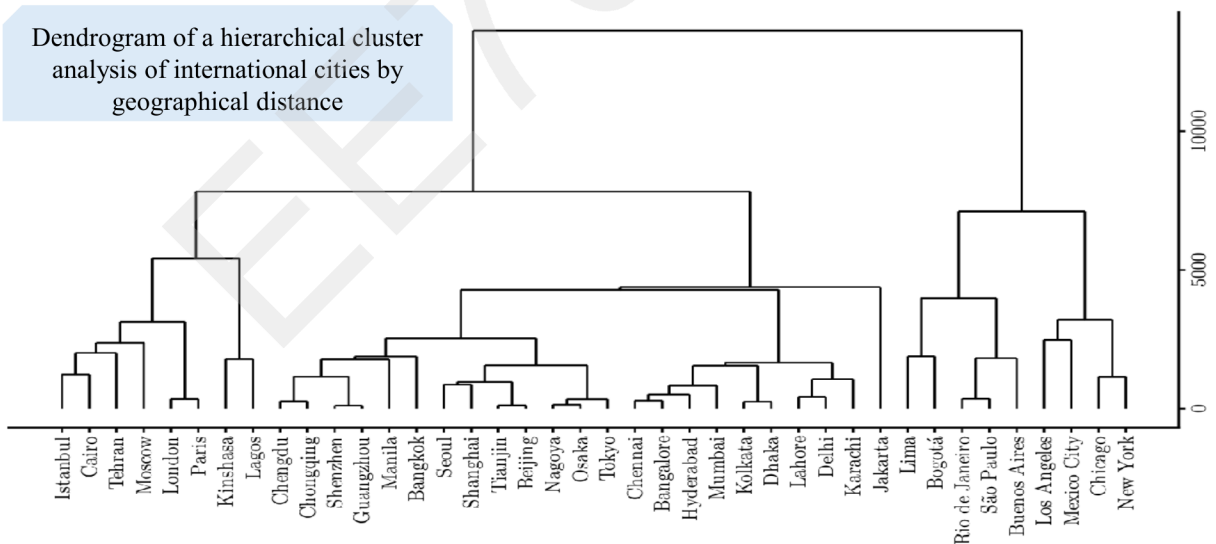
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Hierarchical Cluster Analysis

Dendrogram of a hierarchical cluster analysis of international cities by geographical distance



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